

Cut & Fill Method

16/06/2020

Applicability I

- *Cut & fill stoping is an under-ground mining method.
- *Cut & fill has a single excavation cut (pass /slice) completed and back filled before another cut is made.
- *Cut & fill can be applied for ore body dipping 30 degree to 90 degree.
- *Cut & fill method is best suitable for irregularly shaped deposit and have poor continuity.
- *Cut & fill deposit can be applied to 3 m or wider deposit

Applicability II

- *Cut & Fill stopes depends on gravity to flow broken rock from stope.
- *Chute (ore pass) can be in back fill or in foot wall of stope.
- *Sufficient, suitable filling material should be available.
- *In some mechanized variant, LPDT enters directly in stope & haul out ore out of stope.
- *Some variants of Shrinkage Stoping do need any ore pass or ladder way to lower level through filling.

Mechanized Cut & Fill Stope I

- *Mechanised cut & fill method deploys diesel LHD, LPDT. Drill Jumbo, Site Mixing Explosive truck, rock bolter, truck mounted shotcrete machine, ripper for loose dressing.
- *Machines are not captive in nature, movability of these machines are important to achieve higher machine utilization.
- *To achieve high productivity & lower cost of production, volume of development has been kept to minimum.

Mechanized Cut & Fill Stope II

- *High level of mechanization keeps minimum deployed of men in stope, thus reduces human exposure, man power cost & increases safety.
- *Mechanized method needs higher level of water pumping to ensure faster filling of stopes.
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Mechanized Cut & Fill, Development I

- *A drift is excavated in foot wall of ore body.
- *Foot wall drift is usually 30 to 35 m from ore body.
- *Ore body is accessed by excavating a crosscut from drift.
- *At upper level, ore body is reached by across cut from upper level drift.
- *One or two raises are excavated from lower level to upper level, connecting cross cut from drift.
- *Initial slice (sill level) is excavated from lower level cross cut.

Mechanized Cut & Fill, Development II

- *Development is done by drill Jumbo with 4m drilling round.
- *Size of drive. Cross cut, ramp etc should be able to accommodate the largest machine.
- *Blasting is done with Site Mixing Explosive (SME) mounted on truck.
- *Rock bolting is done with help of “Bolter”.
- *In case, rock formation requires additional support, shot Creating is done with truck mounted machine.
- *Blast drill hole depth can be increased, thus increasing productivity of stope.

Mechanized Cut & Fill, Stoping I

- *Stoping is done with help of uppers drilled by drill jumbo in back of stope.
- *Back of stope provides non stop un-interrupted drilling space & avoids cyclic drilling blasting.
- *Blasting is done with help of SME, mounted on truck.
- *After dressing & rock bolting, broken muck is removed from stope by LHD & LPDT.

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Mechanized Cut & Fill, Stoping II

- *Uppers drill holes are 10 degree inclination to ward free face.
- *Holes should be inclined at in dip direction to reduce over excavation of hang wall & reduction of unbasted ore in foot wall.
- *Filling of stope is done after drilling holes at stope back to reduce maximum open height of stope.
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Mechanized Cut & Fill, Stoping III

- *LPDT travels down from stope to lower level for dumping ore in ore pass.
- *After dumping, LPDT comes back to stope for further loading at stope.
- *Rock bolts are installed by rock bolting machine at HW & back, after face is dressed by “Scaler / Ripper”.
- *Some time rock bolts are also installed at Foot Wall.
- *Once slice is excavated & mucking is completed, stope entry cross cut is heightened for next slice.

Mechanized Cut & Fill, Stoping IV

- *Blasted muck in cross cut is left in cross cut to raise floor level.
- *A barrier is erected at mouth of cross cut, where ever required, to retain sand in side stope at time of filling.
- *After filling of stope, LHD is operated to level floor of cross cut, so that other machine can operate in cross cut.
- *Heightening of cross cut continues to provide access for upper slice till angle becomes 15 degree (maximum trackable gradient of machine).

Mechanized Cut & Fill, Stopping V

- *After crosscut reaching at 15 degree angle, ramp is taken up at 9 degree inclination to make stope entry at elevated position.
- *From elevated location, a ramp down crosscut is excavated to reach stope.
- *This process is repeated till stope reaches upper level.
- *Upper level crown pillar can be recovered based on geomechanically conditions.

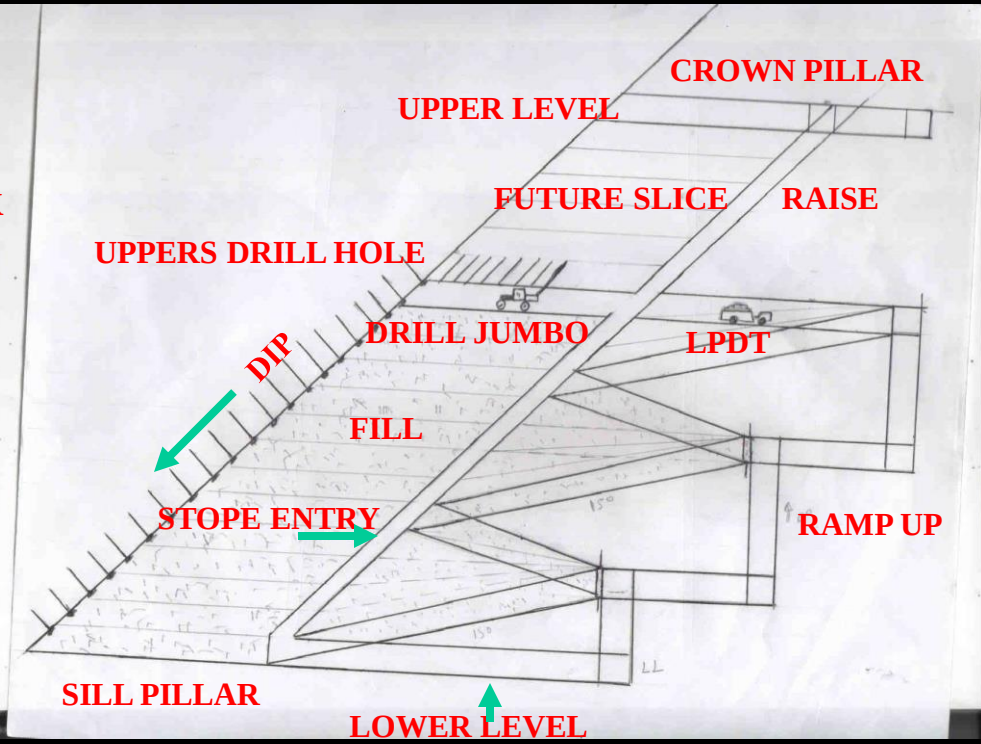
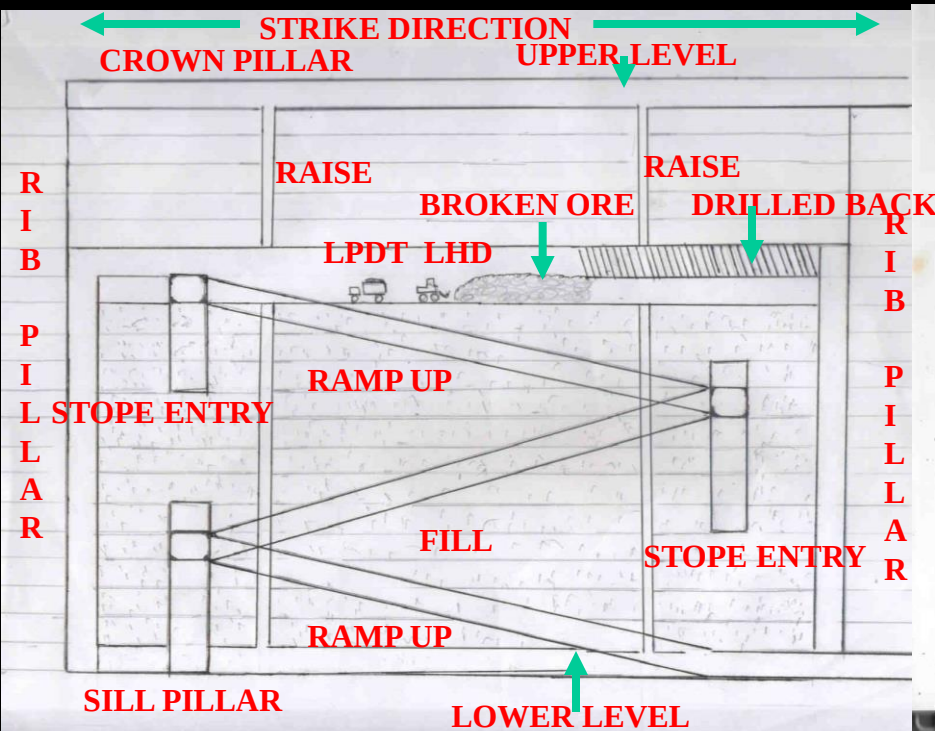
Mechanized Cut & Fill, Ventilation I

- *In mechanized stope, concentration of diesel operated machines are high.
- *To reduce the effect of diesel fume & heat, a suitable booster fan has to be installed on top of raise.
- *This shall ensure flow of air in stope, reduce stope temperature & gas concentration.

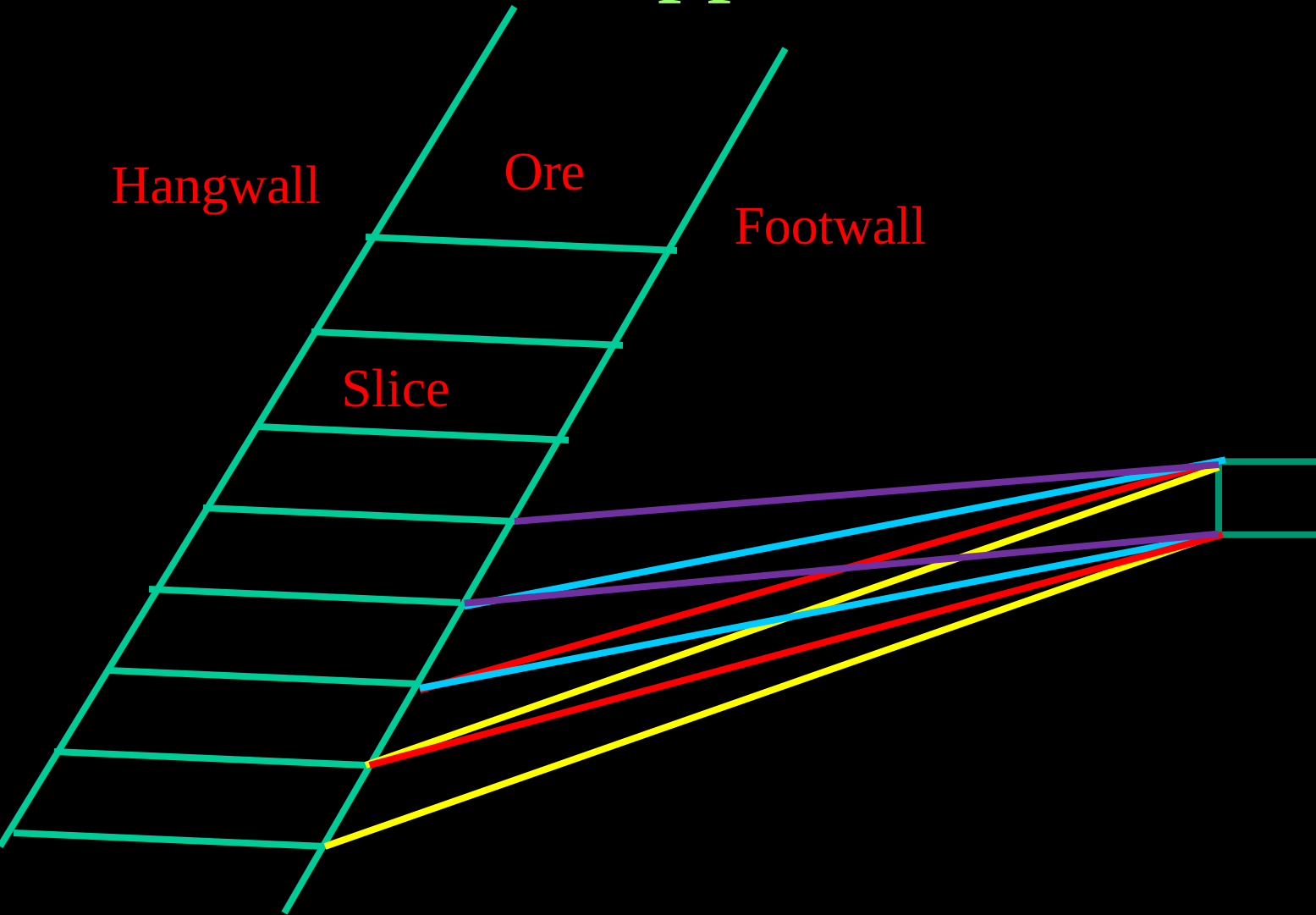
Mechanized Cut & Fill, Safety I

- *Big diesel operated machines has inherent risk of fire & persons getting run over.
- *All machine should be fitted with automatic fire extinguisher to stope fire at beginning it self.
- *Size of human is much smaller in compared to machine being operated in confined space.
- *In this condition, there is all possibility, man gets run over by machine.
- *To prevent this type of accident, presence of man on foot in side stope should be avoided as far as possible.

Mechanised Cut & Fill Stopping* with LHD, LPDT, Drill Jumbo



Cut & Fill Approach to Cut/Slice*



Over Head Cut & Fill

Development I

- *Drives are excavated at lower level & upper level following ore body strike direction.
- *Two raises are taken from lower level to upper level at both limits of stope.
- *Sill level drive is taken in ore body following strike direction.
- *Short raises are excavated from main level to sill level for chute, ladder ways. ventilation passage etc.
- *For wider ore body, connection at different height are made with ramp for entry of LHD / LPDT etc at foot wall.

Over Head Cut & Fill

Drilling I

- *Air leg (jack leg) mounted pneumatic drill machines are used for breast stoping for manual mining.
- *Hole start from 0.6m rod, changing to 1.2m rod and finally with 1.8m rod.
- *Integral drill rods are mostly used.
- *Electro Hydraulic drill jumbos are used for mechanized stopes.
- *Drilling can be upper or breast drilling.
- *For manual stoping, stoper are used for drilling upper hole in to back.

Over Head Cut & Fill

Drilling II

- *Holes are drilled parallel to HW / FW at vertical ore face.
- *Top hole is given slight 2 to 3 degree inclination upward for clear next drill face for breast stoping.
- *For back stoping side holes are drilled slight 2 to 3 degree inclination towards FW & HW.
- *Spacing and burden of holes, drill hole pattern depends on (1) type of ore (2) ore handling arrangement at stope, chute, ore pass and hoisting.
- *Any ore left on periphery of stope from last cut blast is to be stripped before next round is blasted.

Over Head Cut & Fill Breast Stope I

- *Breast stoping is a method of extracting a horizontal block (slice) of ore by drilling horizontal holes.
- *Mining progress up wards from a lower level to upper level.
- *Horizontal holes (1.5m to 4.6 m) are drilled in face (breast).
- *After blasting ore face, new exposed face and back are dressed (scaled).
- *Broken ore is removed by LHD, slusher etc in stope.
- *Ore is dumped in to chute or loaded directly to LPDT.

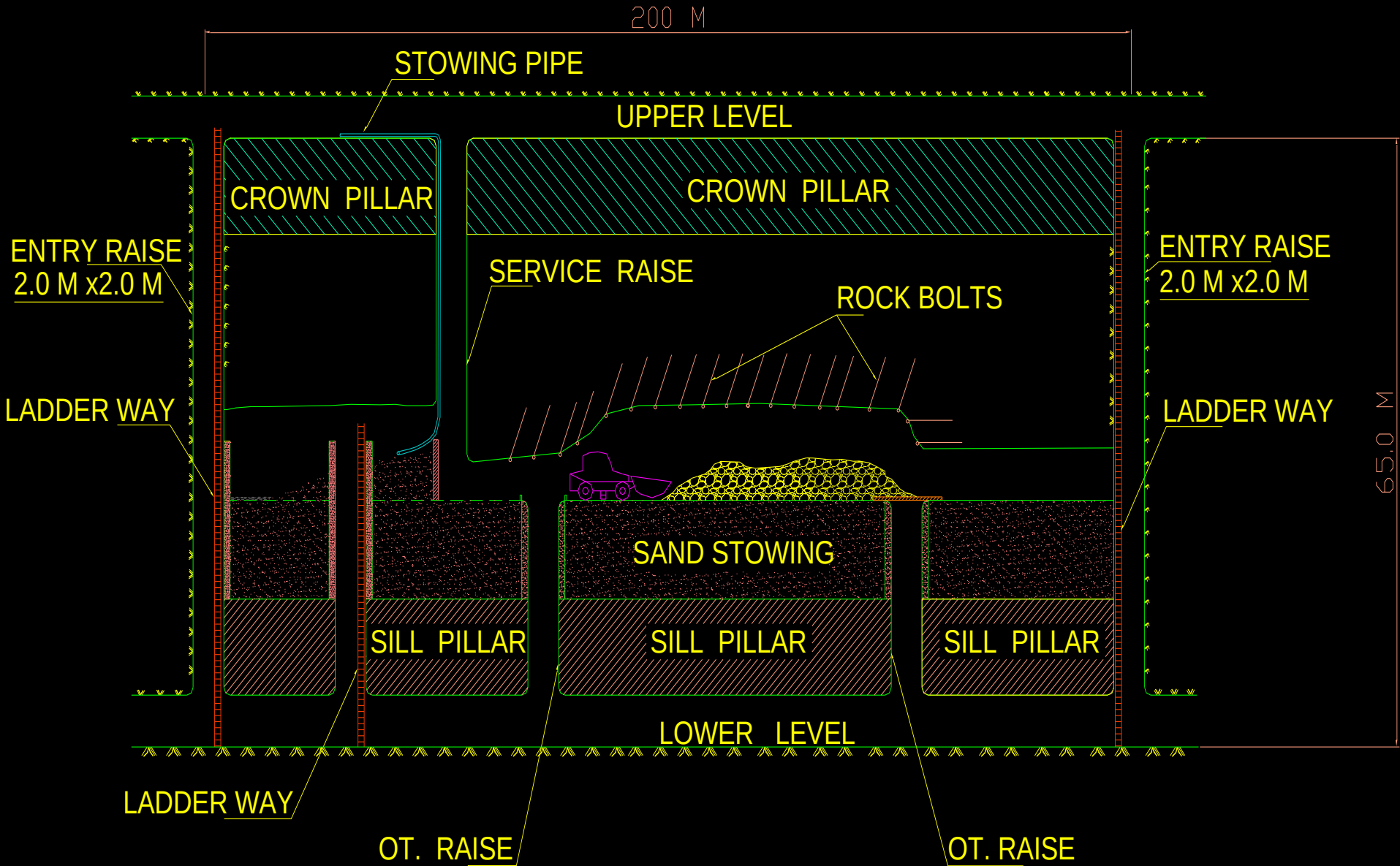
Over Head Cut & Fill Breast Stope II

- *Broken ore is dumped in to chute (ore pass) for transferring to lower level.
- *Ore passes are placed at 5 to 150m distance.
- *Distance in between ore pass is decided primarily by allowable travel distance of loading machine.
- *Slice is extended from one ore pass to another ore pass.
- *After completing a horizontal cut, ore passes are extended up wards to height of ore removed and filled.
- *Another cut is made above back fill.
- *Cutting continues till cut reaches crown pillar.

Over Head Cut & Fill Breast Stope III

- *Crown pillar can be mined by filling last cut.
- *Drive at upper level can be maintained over fill.
- *Stress on crown pillar increases at end of stoping operation.
- *Raises are excavated from lower level to upper level are used as service, ventilation and communication purposes.

Over Head Cut & Fill Breast Stope



Non Mechanized Breast Stope

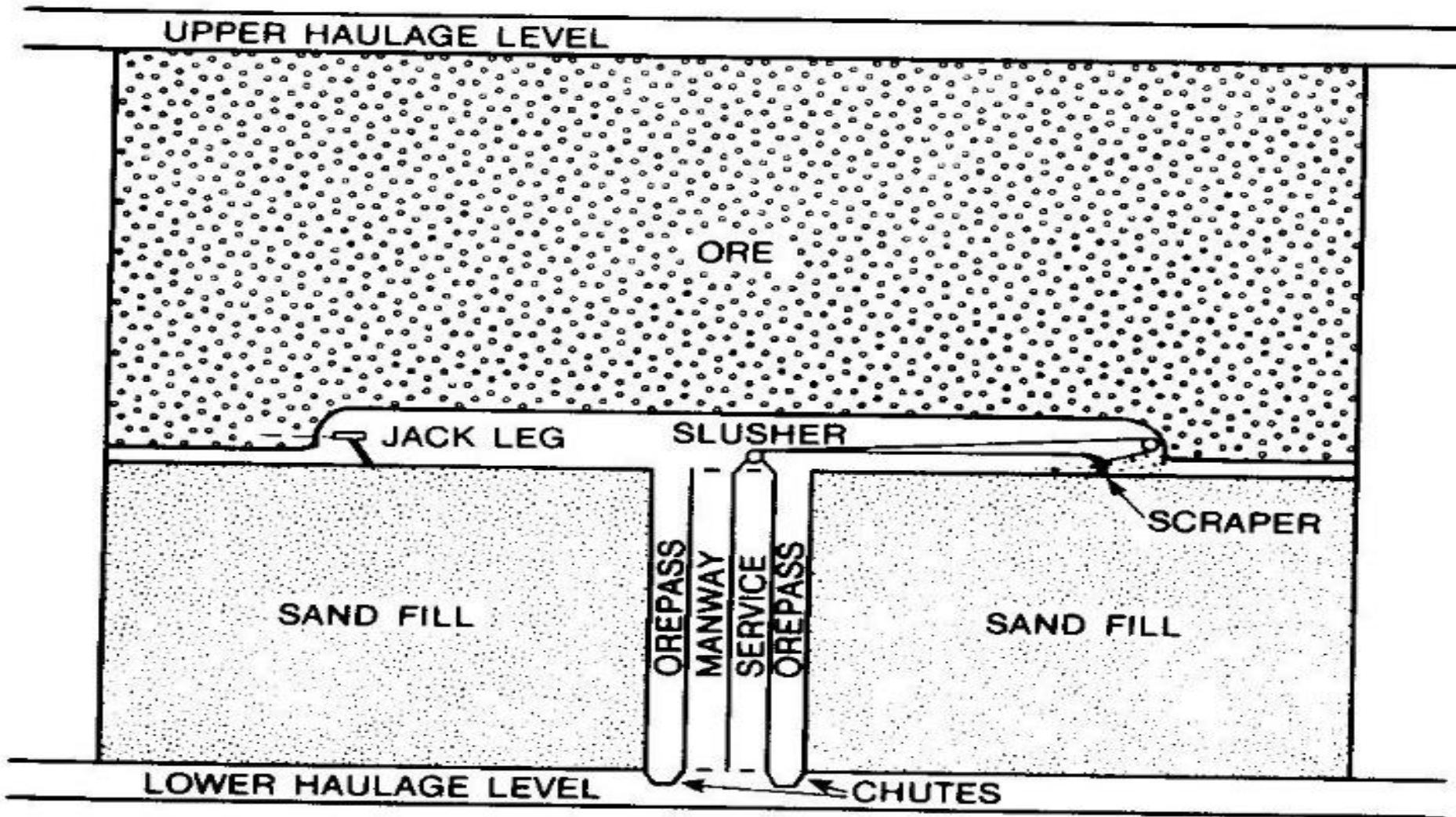
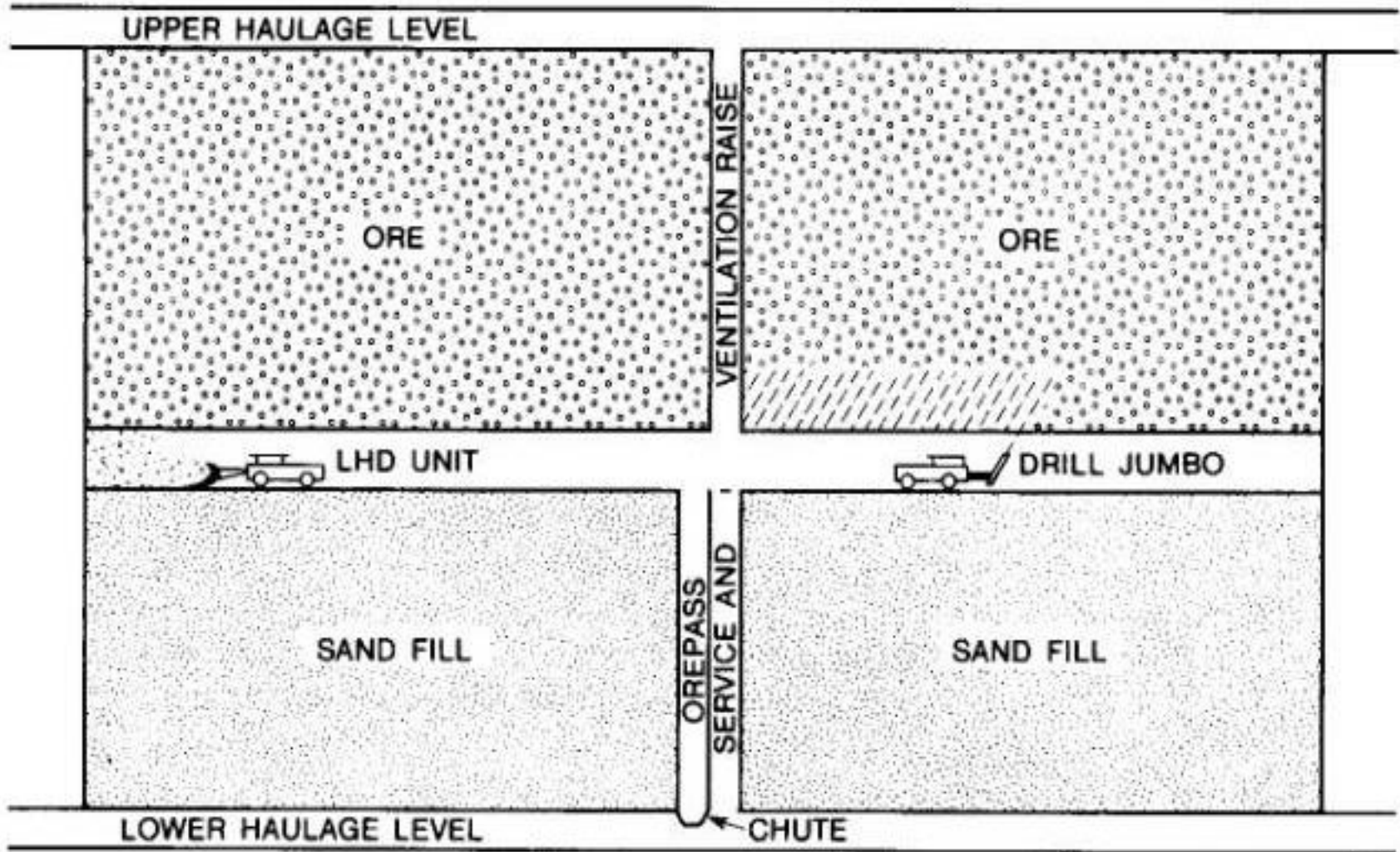


Fig. 19.2.1. Longitudinal section of a typical non-mechanized breast stope.

Mechanized Breast Stope

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Over Head Cut & Fill Breast Stope

Post Pillar I

- *Post pillar method is identical to over head cut &fill (breast) method with addition of vertical pillar in stopes.
- *Post pillar stoping is used to mine wider ore body
- *Base of vertical pillars are formed on sill pillar at times of first cut.
- *Pillar size and positions are decided by geo - mechanical study of ore body and stope configuration.
- *Waste rock portions are also left as pillar as far practical.

Over Head Cut & Fill Breast Stope

Post Pillar II

- *After some slices / cut, pillars get merged in to hang wall.
- *Subsequent, new pillars are initiated from foot wall.

Over Head Cut & Fill Breast Stope

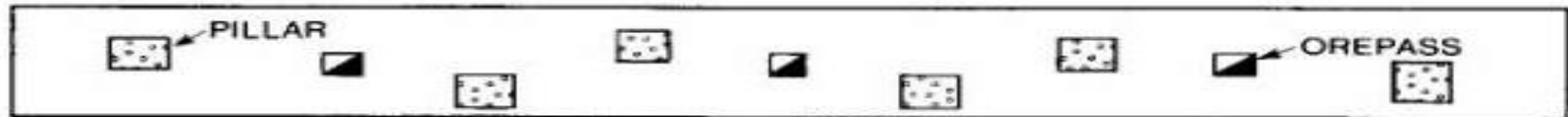
Post Pillar III

- *Post pillar stoping is applicable to wider ore bodies.
- *Competency of ore should be good to prevent failure of pillar, back and HW with pillar support.
- *Horizontal slice is cut after leaving vertical pillar for supporting back.
- *First horizontal slice is filled keeping the pillar intact.
- *As mining progresses up wards, tall slender pillars are prevented from buckling by fill material.

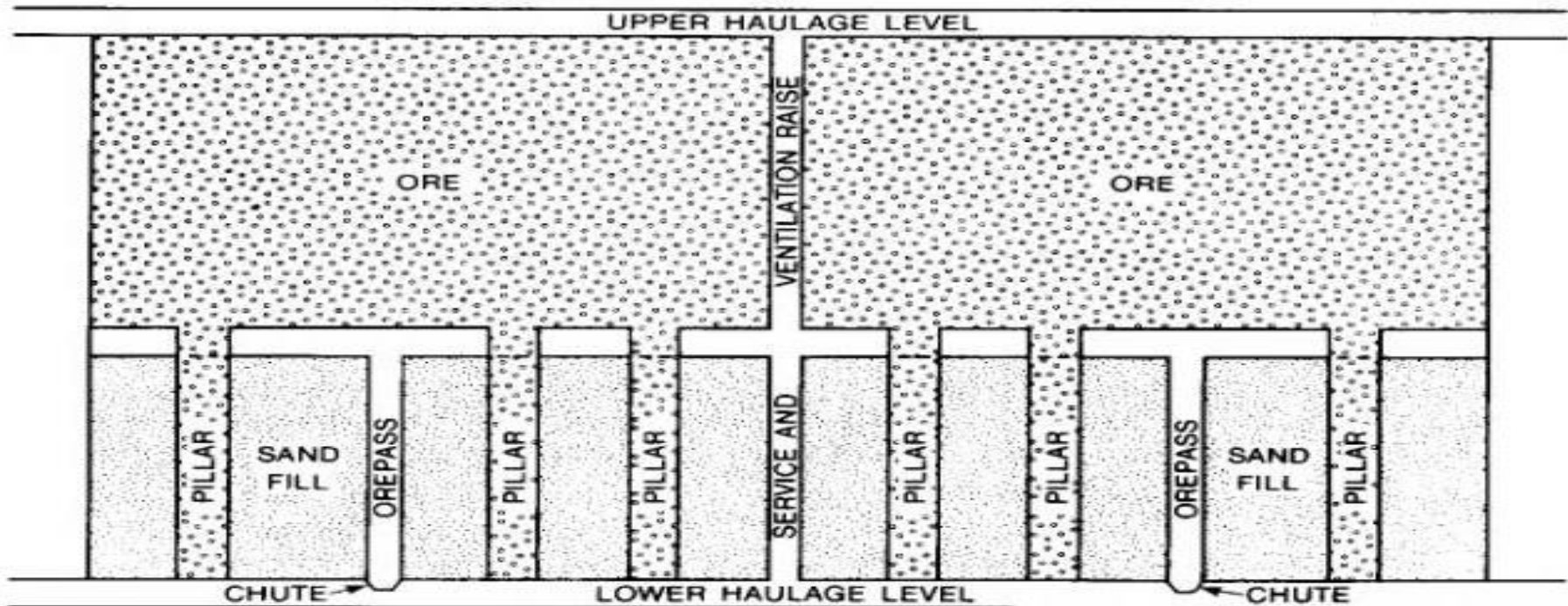
Post Pillar Stopping

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PLAN



LONGITUDINAL SECTION

Over Head Cut & Fill Back Stoping I

- *Back Stoping method is identical to over head cut & fill (breast) method with condition that breast holes are replaced by upper blasting holes.
- *Horizontal slices of 1-3 m thickness is removed by drilling holes at steep angle (50-70 degree)
- *Blast holes are drilled from one ore pass to another ore pass.
- *Holes are loaded with explosive and blasted at one time.
- *Back stoping is particularly suitable for steeply dipping , narrow vein ore bodies.

Over Head Cut & Fill Back Stopping II

- *Back Stopping method needs enough open space between backfilled surface and stope back to drill vertical or steeply inclined blast holes in up ward direction.
- *Drilling Jumbo can be used for drilling uppers before back filling the stope.
- *Back stopping is not preferred, when back (roof of stope) is weak & not sufficiently competent.
- *Blasting shock from upper holes travels up wards.

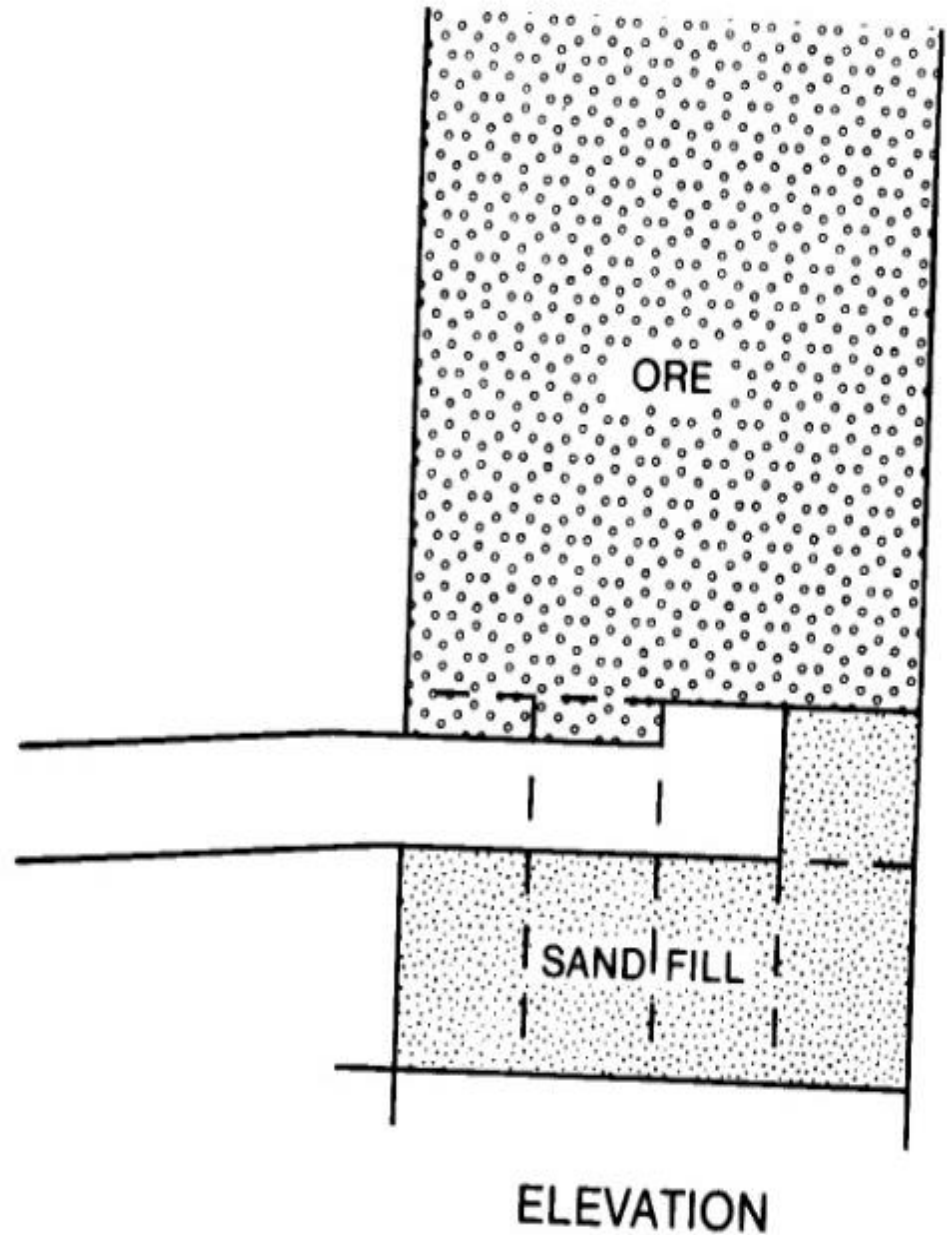
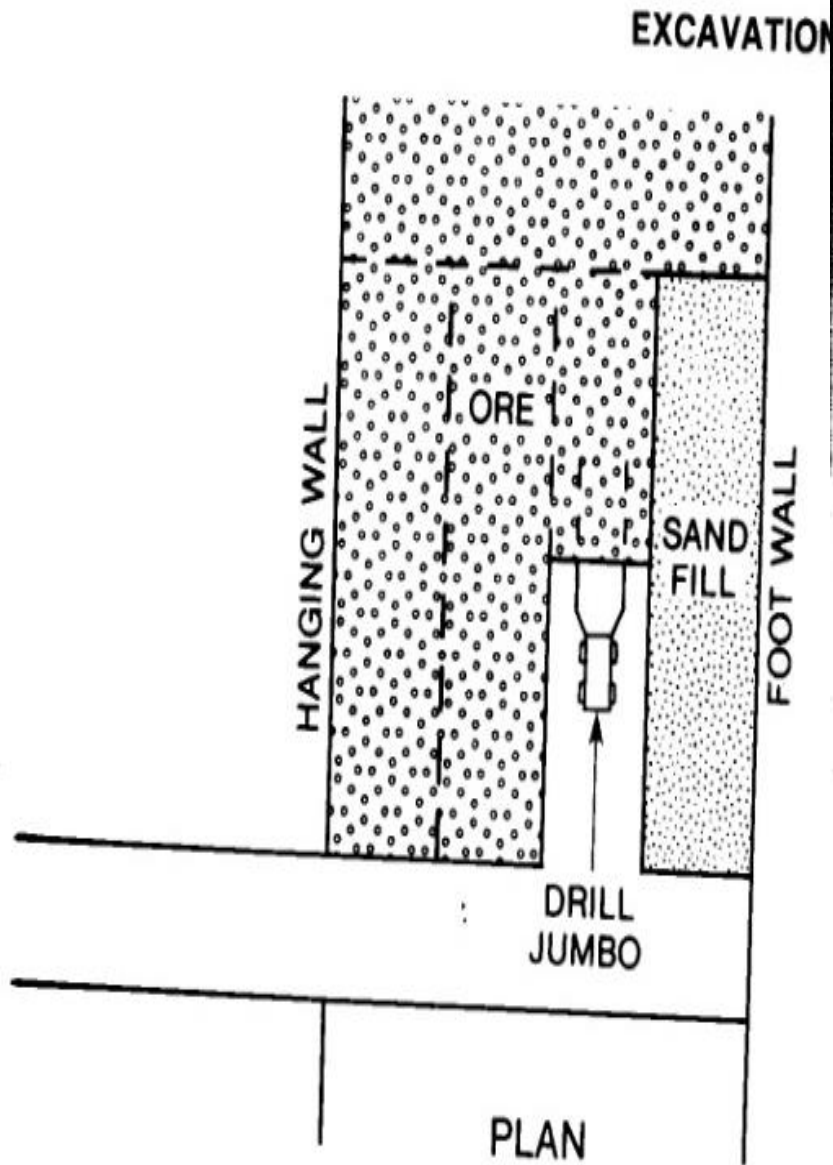
Over Head Cut & Fill, Drift & Fill I

- *Drift & fill stoping is used to mine wide ore bodies (more than 6 m in width).
- *Competency of ore is moderate to poor.
- *Mining block is divided in to series of drifts parallel to strike with vertical walls.
- *Mining starts by stoping in to series of parallel drifts with vertical walls.
- *Stopes area is then filled with cemented sand fill.
- *Cemented back fill supports back of stope for next parallel drift.

Over Head Cut & Fill, Drift & Fill II

- *Another drift is excavated adjacent to back filled drift.
- *This sequence is repeated until entire width of ore is mined out.

Drift & Fill Stopping



Over Head Cut & Fill

Transverse Stope & Pillar I

- *Ore body is mined by dividing in into transverse stopes and intervening rib pillars.
- *During mining of stopes, rib pillars serve as abutment for back and provide some over all support to hang wall.
- *Stopes are filled with cemented sand fill and rib pillars provide partial support.
- *Rib pillars offer resistance to closer between hanging wall and foot wall.
- *Extraction of rib pillars are done by under cut and fill method.

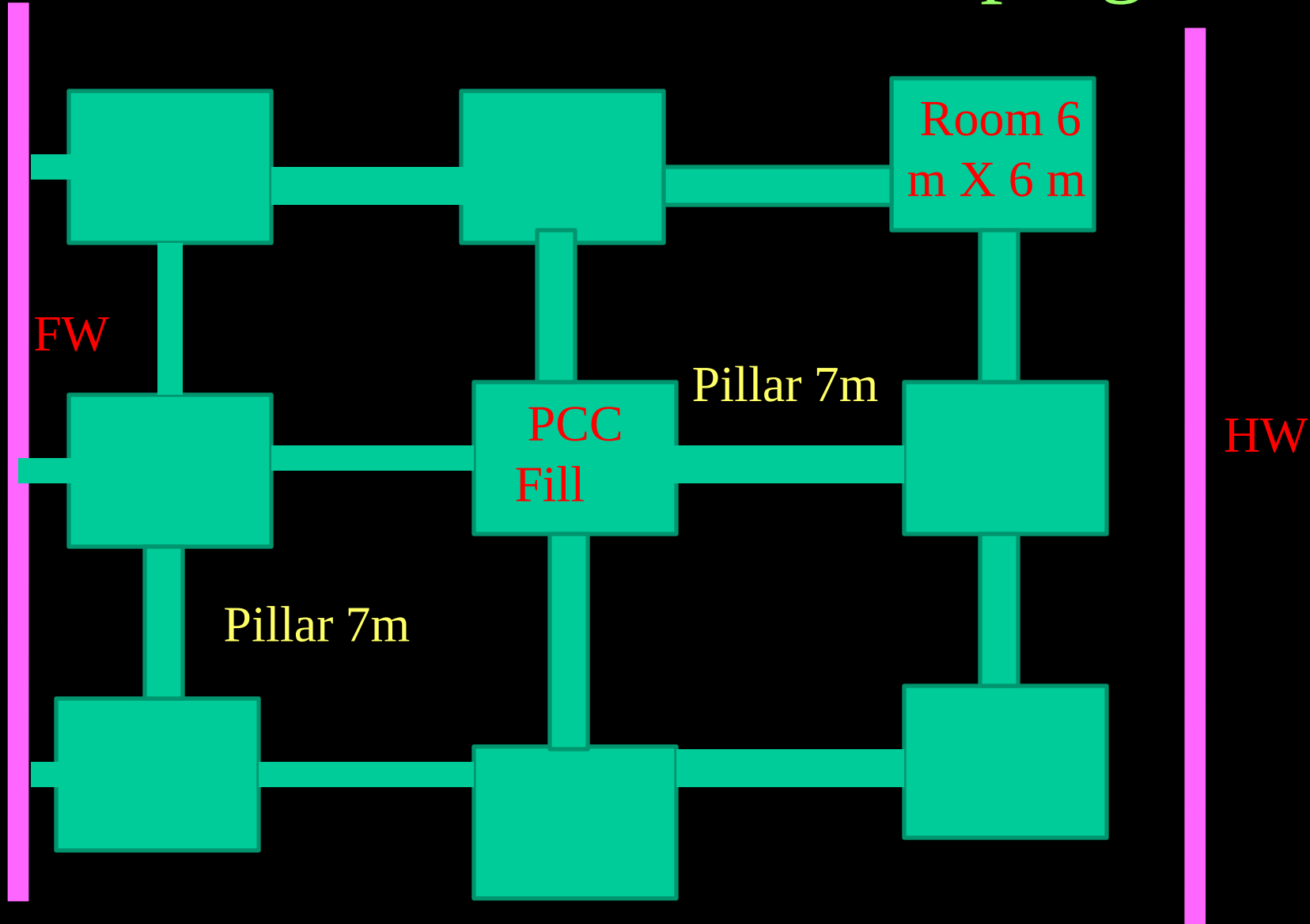
Over Head Cut & Fill

Concrete Pillar Stopping I

- *Mining block is divided 6m by 6m parallel rooms with vertical wall separated width 7 m between rooms.
- *6 m rooms are stoped first and then rooms are back filled with concrete.
- *Then 7m wide ore pillars are stoped and they are filled with classified tailing and natural gravels.
- *Finland, Outokumpu Company, Keretti Mine, Cu Zn

Over Head Cut & Fill

Concrete Pillar Stopping I



Over Head Cut & Fill Support

Rock Bolt I

- *Rock bolt is popular as support because they are (1) in expansive (2) easy to install (3) efficient (4) does not reduce size of opening (5) can be combined with any other support.
- *Rock bolts are installed in a pattern after each blast.
- *Immediate installation of rock bolt minimizes extend of unsupported back that man must work under or near.

Over Head Cut & Fill Support

Rock Bolt II

- *Rock bolt in muck pile is a nuisance for next cut as well as in mucking operation.
- *Rock bolts are to be removed from broken ore /rock.

Over Head Cut & Fill Support

Resin Grouted Rock Bolt I

- *This rock bolt provides instant support.
- *Grouting length can be varied.
- *Resin cartridge are clean and easy to handle.
- *Bolts can be plain rebar, threaded rebar, cable.
- *Resin is costly, proper mixing of resin and hardener takes place at certain condition.
- *Short self life of resin cartridge.

Over Head Cut & Fill Support

Cement Grouted Rock Bolt I

- *Cement grouted bolts have good reinforcement properties.
- *They are protected from corrosion.
- *Cement is in-expensive and readily available.
- *Bolts has long setting time.
- *A mixture and grout pump is required to install rock bolts.
- *Not very convenient for installing few grouted bolts.

Over Head Cut & Fill Support

Other Supports I

- *Steel crib set made of hollow square tubes and assembled at site with the help of clamps, nut and bolts.
- *Concrete crib set made with precast concrete square section pillar.
- *Both crib sets are jammed with wooden wedges.
- *Some time wooden wedges are placed in between two wooden crib sets.
- *Cable bolt is used when prominent joint plane is present at depth, which can not be reached by rock bolts.

Over Head Cut & Fill Support

Other Supports II

- *Fill in cut & fill method provides maximum permanent support to hanging wall and foot wall.
- *Post pillar provides permanent support to back and hanging wall.
- *Verticality of Post pillar support is most important for providing support to back and hanging wall.

Over Head Cut & Fill Support

Expansion Shell Rock Bolt I

- *Expansion shell rock bolts are point anchored bolts.
- *Expansion shell rock bolt tends to loosen strength after installation.
- *Expansion shell rock bolt needs to be checked and if necessary, needs to be tightened
- *Frequency of checking & tightening depends on ground condition and proximity of blasting.
- *After removal of ore, rock bolts becomes inaccessible and checking & re-torqueing becomes impossible creating an unsafe condition.

Over Head Cut & Fill Support

Friction Stabilizer Rock Bolt I

- *This rock bolts are very fast and easy to install.
- *Holding force acts over full contact length of bolt.
- *Bolt has ability to yield, while retaining full holding force.
- *Anchorage increase with time.
- *Plate loads are maintained.
- *Full contact around circumference of hole provides a greater resistance to displacement of rock along shear and bedding plane.

Over Head Cut & Fill Support

Friction Stabilizer Rock Bolt II

*Does not have holding power of resin grouted bolts in hard and fractured ground.

Over Head Cut & Fill Support

Timber I

- *Timber supports like stall, chock mate, post, crib set are placed at hang wall, back, foot wall as per requirement.
- *Timber supports is mainly temporary but can be permanent.
- *Timber stull is used after each blast, where ever required to increase safety of miner.
- *Timber stulls are installed before mucking . Stull cap is lightly installed with a heading (wedge) against FW & HW.

Over Head Cut & Fill Support

Timber II

- *Heading absorbers wall closer with out causing cap to be crushed in compensation.
- *Wood fiber in heading are at right angle to pressure.
- *Wood is compressible when confined at right angles to the fibers.
- *Cap (stull end) is protected until wall closures has compressed heading to full extend.

Under Hand Cut & Fill Applicability

- *H/W, F/W or ore vein is very unconsolidated to permit safe mining by over cut and fill method.
- *Ore bodies with changing width.
- *Ore bodies with changing dip vertical to flat lying.
- *For recovering pillars left by other method of working.
- *Method eliminates problem of removing pillar of decreasing area which supports horizontal stress.
- *In deep mining, this method reduces rock burst, specially when filling is done with concrete.

Under Hand Cut & Fill Method I

- *Under Hand / Under Cut & filling is a method of extracting a block of ore by mining successive slices (cut) 2 to 5 m in height from top down and filling each successive cut with cemented sand fill.
- *Ore body is developed by developing levels at foot wall at 40 to 50 m interval.
- *Extraction cross cut are driven through vein (ore) at 30 to 90m interval.
- *Ore pass raises are driven or bored in ore body or wall rock between upper and lower crosscut.
- *In mechanized mines ramps are driven for access to stopes and ore pass.

Under Hand Cut & Fill Method II

- *Initial floor is established on upper level.
- *Floor of level is mined out by driving conventional drift and supported by rock bolts and mat.
- *Ore is removed by slusher / LHD (for mechanized mines), through ore pass to level below.
- *After completing initial cut, timber, wire fabric mat are placed on floor of cut.
- *Opening is filled tightly with cemented sand fill.
- *Access to mining area on next cut is maintained by downward man way and timber slide for material transport or ramp down wards.

Under Hand Cut & Fill Method III

- *Slice below cemented fill is mined below mat.
- *Sand filled roof / back with mat is supported by timber post (stull) if necessary.
- *Undercut method is used at great depth by using concrete as filling material (Kolar, Champion Riff).
- *Cemented sand filled roof / back can also be supported by wire fabric (in place of post).
- *Wire Fabric is bolted at both side wall (HW & FW).
- *Fill placed in each cut must be competent.
- *There must be adequate cemented sand fill material available at stope with dependable sand plant and distribution system.

Under Hand Cut & Fill Development I

Non Mechanized stope

- *Lateral (drive parallel to strike direction) are taken in foot wall of ore body.
- *Extraction cross cuts are driven in to ore body at 30m to 100m interval depending up on length of vein and efficient slushing / loader travel distance for mine.
- *Raises for ore passes are driven in vein /FW up to upper level.
- *Raises can be bored from lower level to upper level in ore or in FW of deposit.
- *Raises and ore passes are taken in FW of lode when ore body is not competent.

Under Hand Cut & Fill Development II

Non Mechanized stope

*Loading chutes are then installed at bottom of ore pass raise.

Under Hand Cut & Fill Development I

Mechanized stope

- *A main ramp decline is driven in wall rock (FW) from upper level to lower level.
- *Decline are driven at some pre determined distance in wall rock (10 to 15m).
- *Cross cut access ramps are then driven from main ramp in to ore body.

Under Hand Cut & Fill Access I

- *Access to stope is via winzes that are carried down as mining progress.
- *Access to stope is maintained by ramp for mechanized stope.
- *Access can also be provided by raise excavated at the time of development.
- *Three compartment raises can also be provided as access, ore chutes, service entry (slides).

Under Hand Cut & Fill Drilling I

- *Drilling is done with Jack legs (Jack hammer with pusher leg) or drill jumbo.
- *Before drilling face is dressed (barred down).
- *Under side of mat is checked to remove any loose sand fill.
- *Top holes are drilled flat and a minimum distance (0.5 to 1m) below mat in order to prevent damage to mat when round is blasted.
- *Lifter holes are drilled at bottom of drive, so that unnecessary blasting to be avoided at floor for setting stull post or stringer (flat wood supporting stall at bottom or top)

Under Hand Cut & Fill Drilling II

- *Floor should be as smooth as possible for easy mucking.
- *Normally, 1.8m round is blasted with burn cut.
- *While blasting, care has to be taken, not to damage mat over head.

Under Hand Cut & Fill

Ground Support I

- *Temporary ground support is provided by post / rock bolts.
- *With progress of mining, round, wood posts are placed under flat wood support (stringer) of previous cut.
- *Rock bolts are also be used to supplement support (post) .
- *In narrow vein, mostly rock bolts are used to provide temporary support till mat is constructed and cemented sand fill is poured.

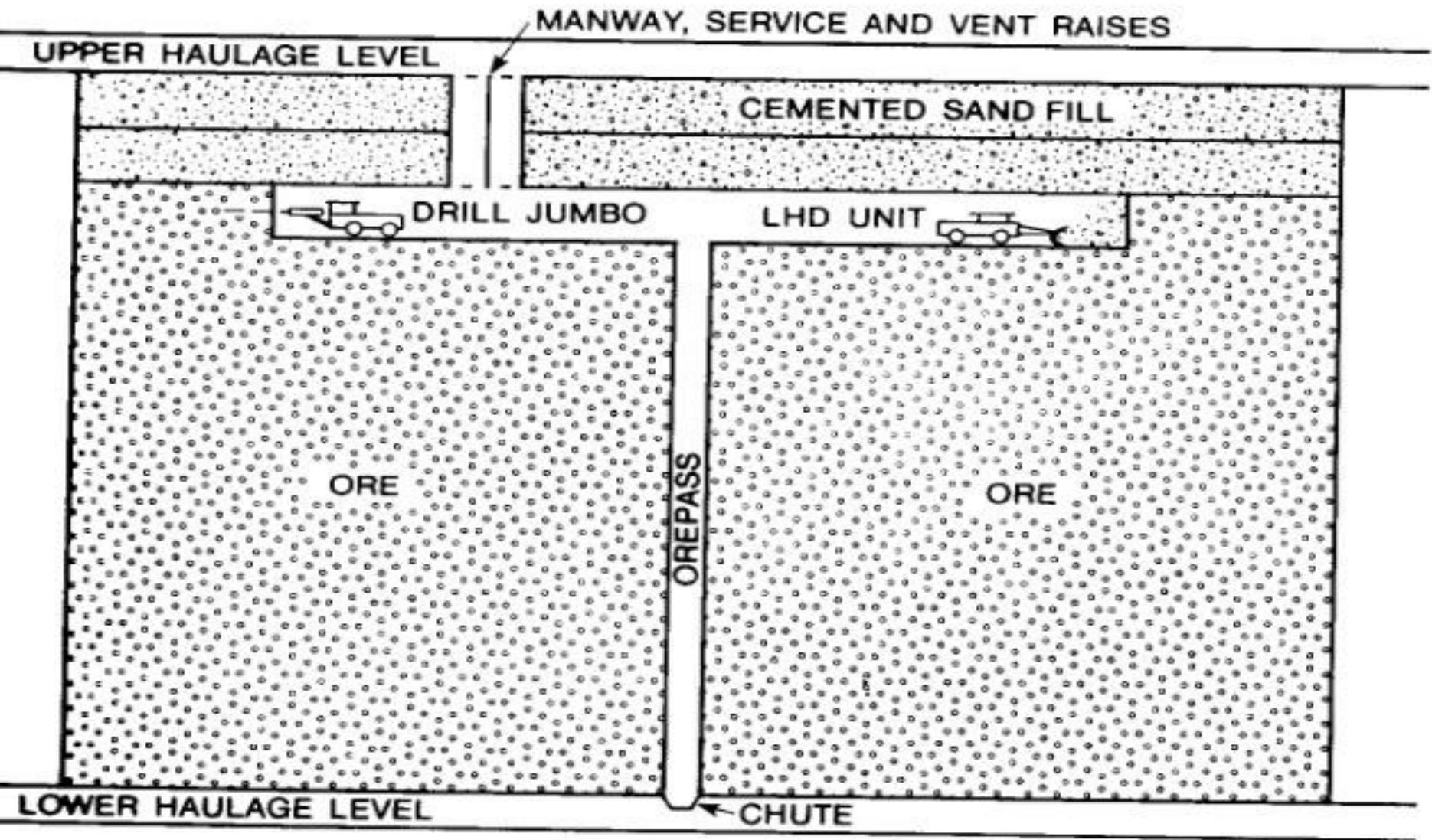
Under Hand Cut & Fill

Ground Support II

- *Wire fabric is spread in two ribs (HW & FW) and floor while is held in place with rock bolts.
- *Cemented sand fill supports major portion of load.
- *Post and mat have double function as they carry weight not supported by sand fill.
- *Post and rock bolts provide pronounced setting of fill mass.

Cut & Fill Under Cut

EXCAVATION TECHNIQUES



Cut & Fill Blasting I

- *Every hole is loaded with a primary cartridge loaded with electric or non electric detonator.
- *Long delay detonator is preferred to reduce ground vibration.
- *Holes are loaded with explosive before blasting.

Cut & Fill Ore Removal I

- *Ore is removed from non mechanized stope by means of double drum slusher (scrapper).
- *Slusher can be electric or pneumatic powered.
- *Ore is scrapped to stope ore pass where it drops to level below.
- *In mechanized stope ore is removed from stope by diesel or electric power LHD.
- *Ore is dumped at stope ore pass for handling at lower ore level or ore is directly transported to main ore pass.

Cut & Fill Ore Removal II

- *As soon as stope is mined out, all ore must be removed from stope. This is called clean out cycle.
- *In stope using slusher, scrapper (hoe) should be played back and froth across muck pile to keep muck level low.
- *Trenching or winrowing should be avoided.
- *Clean out should progress from back end of stope to wards ore pass.
- *Slusher should take minimum amount of sand with hoe (scrapper) to ore pass along with ore, where cemented filling is not used.

Cut & Fill Ore Removal III

- *LHD can do mucking, cleaning stope much faster.
- *LHD operator should be care full not to dig too far into sand if cemented fill is not used.
- *Remaining ore in wall FW & HW to be taken out at end of clean out cycle as muck level in stope is lowered.
- *Whenever necessary for safety or to prevent dilution, wall shall be supported by stull or rock bolts.

Cut & Fill Ore Pass I

- *Ore pass can be unlined when they are located in competent country rock.
- *Ore pass in back fill are to be constructed by wood, concrete or steel plate ring.
- *Some time ore passes are lined by steel fiber enforced shot-crete.
- *Choice of lining depends on type of rock, length of ore pass, estimated tonnage that will flow through chutes & life of chute.
- *Unlined ore pass in hard tough country rock stands up under constant flow or ore.

Cut & Fill Ore Pass II

- *Bottom of ore pass at haulage level contain a loading chute, that is lined up with steel plate and equipped with pneumatically operated gate.
- *Ore is loaded from chute gate in to truck (LPDT) or car (train) for dumping ore in main ore pass near hoisting shaft.
- *Ore may be dropped to an enlarged area beneath ore pass.
- *Ore then loaded in to mine car (train) or LPDT (truck) using LHD.
- *Ore can be directly hauled by LHD to main ore pass.

Cut & Fill Ventilation I

- *Fresh air is taken in mine through down cast shaft.
- *Air is splited to flow through active work area.
- *Return air is exhausted through raise, laterals and finally through up cast shaft.
- * Wet bulb temperature at work place shall be 30.5 degree C (maximum), dust level below 0.8mg/m cu, CO, 50 ppm, Nox 5 ppm,
- *Ventilation air requirement and net work is to be designed for each individual mines, full filling the statutory and other requirement.

Cut & Fill Ventilation II

- *Suitable auxiliary and booster fans are to be considered for fulfilling local ventilation requirement.
- *For deep mine, air conditioning, cooling system has to be designed as per requirement.

Advantages I

- *Among underground mining method, where man is exposed inside active stope, cut & fill method offers very high level of safety.
- *Mine fill improves safety and wall support by limiting working height of stope.
- *Back fill provides a working floor at a convenient elevation for mining activities.
- *Very irregular ore body can be mined precisely.
- *Greater selectivity results in a higher grade ore product.
- *Wall supporting function of back filling aids in prevention of back / hang wall failure in stope.

Advantages II

- *Prevention of back / hang wall failure reduces chances of dilution in ore, possibility of air blast in stope, subsidence at surface.
- *In case of cemented back fill, dilution by mixing with fill is reduced.
- *In under cut method, dilution does not take place by mixing with fill.
- *Lower rock quality designation of ore and poor quality of enclosing rock (HW and FW) can be mined by cut & fill method.

Advantages III

- *waste rock can be dumped in side empty stope, this avoiding hoisting & dumping of waste rock on surface.
- *In cut & fill mining, there is no possibility of surface subsidence, possibility of air blast & stope collapse is remote.

Disadvantages II

- *Dilution takes place because ore gets mixed up with back fill at upper hand method.
- *In waste filled stopes, fine particles of ore gets lost in void of waste rock.
- *In under cut method, dilution does takes place by fallen back fill from stope back.
- *Over hand cut & fill method, when advancing from lower level to mined out upper level, stress concentration on left out sill & crown pillar goes very high.
- *High humidity level in stopes.
- *High pumping cost.

Disadvantages III

- *Frequent cleaning of drain & main sump.
- *Cut & fill method of working is not bulk production method.
- *Cutting cycle only produces ore , stope preparation & filling cycle is non productive.
- *Cut & Fill method of working is costly method.

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Reference

*SME Mining Engineering Hand Book, 2nd Edition,
page 1743 to 1748